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**B.Tech. Degree II Semester Regular and I Semester Supplementary
Examination May 2016**

IT/CS/EC/CE/EE/ME/SE 1101 A / 1201 B CALCULUS
(2015 Scheme)

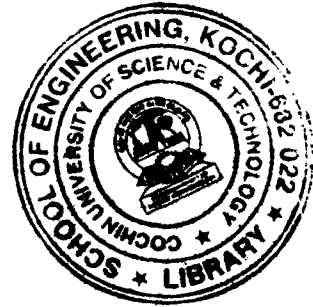
Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Solve $\frac{dy}{dx} = e^{2x-y} + x^3 e^{-y}$.
- (b) Solve $\frac{d^2 y}{dx^2} + 25y = \cos 4x$.
- (c) Solve $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$.
- (d) Test the convergence of the series $\sum_{n=0}^{\infty} \left(\frac{1}{2^n} \right)$.
- (e) Explain the Cauchy's root test and give one example.
- (f) Obtain the Maclaurin series expansion of $f(x) = \sin x$.
- (g) Find $\frac{\partial^3 u}{\partial x \partial y \partial z}$, if $u = e^{x^2+y^2+z^2}$.
- (h) Find the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ of $u = x \sin y$; $v = y \sin x$.
- (i) Find the area bounded by the parabolic arc $\sqrt{x} + \sqrt{y} = \sqrt{c}$, $c > 0$ and the coordinate axes.
- (j) Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2) dx dy$.



PART B

(4 × 10 = 40)

- II. (a) Solve $\frac{dy}{dx} = \frac{y-x}{y-x+2}$.
- (b) The slope at any points (x,y) of a curve is $1 + \frac{y}{x}$. If the curve passes through (1,1) find the equation of the curve.
- (c) Solve $4y^2 p^2 + 2pxy(3+1) + 3x^3 = 0$, where $p = \frac{dy}{dx}$.

OR

(P.T.O.)

III. (a) Solve $\frac{dy}{dx} + 2y = e^x (3 \sin 2x + 2 \cos 2x)$.

(b) Solve $4x^2 \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + 3y = 0$.

(c) Solve the simultaneous equations

$$\frac{dy}{dt} + 5y - 2x = t$$

$$\frac{dx}{dt} + 2y + x = 0$$

IV. (a) Explain Raabe's test for testing the convergence of an infinite series and using it test the convergence of the series $\sum_{n=1}^{\infty} \left[\frac{4.7 \dots (3n+1)}{1.2 \dots n} \right] x^n$.

(b) Test for the uniform convergence of the series

$$\frac{\sin x}{1^2} + \frac{\sin 2x}{2^2} + \frac{\sin 3x}{3^2} + \frac{\sin 4x}{4^2} + \dots$$

OR

V. (a) Discuss the convergence of the series

$$1 + \frac{x}{2} + \frac{x^2}{5} + \dots + \frac{x^n}{n^2 + 1} + \dots$$

(b) Obtain power series solution in powers of x for the following differential

$$\text{equation: } (x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0, \quad y(0) = 4 \text{ and } y'(0) = 6.$$

VI. (a) If $u = x^3 y^2 \sin^{-1}(y/x)$ show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 5u.$$

(b) Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube.

OR

VII. (a) If $V = u^2 v$ and $u = e^{x^2 - y^2}$, $v = \sin(xy^2)$ find $\frac{dv}{dx}$ and $\frac{dv}{dy}$.

(b) Given $u = 2axy$, $v = a(x^2 - y^2)$ where $x = r \cos \theta$, $y = r \sin \theta$, find $\frac{\partial(u, v)}{\partial(r, \theta)}$.

VIII. (a) Find the total area between the cubic $y = 2x^3 - 3x^2 - 12x$, x -axis and its maximum and minimum ordinates.

(b) Evaluate $\iint_R (x+y)^2 dx dy$, where R is the parallelogram in the xy -plane with vertices $(1, 0)$, $(3, 1)$, $(2, 2)$, $(0, 1)$ using the transformation $u = x + y$ and $v = x - 2y$.

OR

IX. (a) What is the volume generated by revolving the area enclosed by the loop of the curve $y^4 = x(4-x)$ about x -axis?

(b) Find the mass, centroid of the tetrahedron bounded by the coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. Given that the substance has constant density ρ .